SKTC 1623

Thermodynamics

Chapter 3: Properties of Pure Substances

Dr. Nur Hidayah Zainan

3.1 Pure Substance

- Pure substance is a substance that has a fixed chemical composition throughout.
  - Water, nitrogen, helium, and carbon dioxide etc., are all pure substances.
- A mixture of various chemical elements or compounds also qualifies as a pure substance as long as the mixture is homogeneous. E.g air.
- Mixture of two or more phases of a pure substance is still pure if the chemical composition of all phases is the same. E.g water

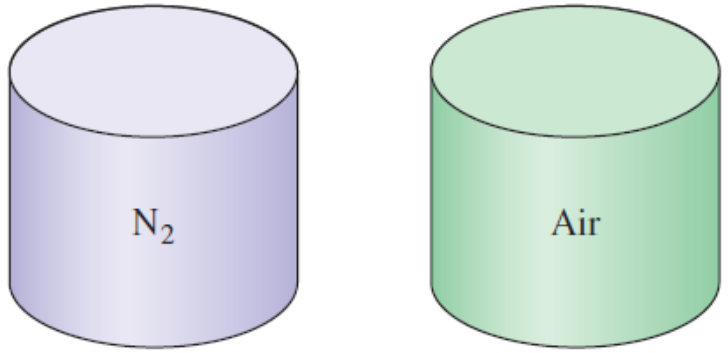


FIGURE 3–1
Nitrogen and gaseous air are pure substances.

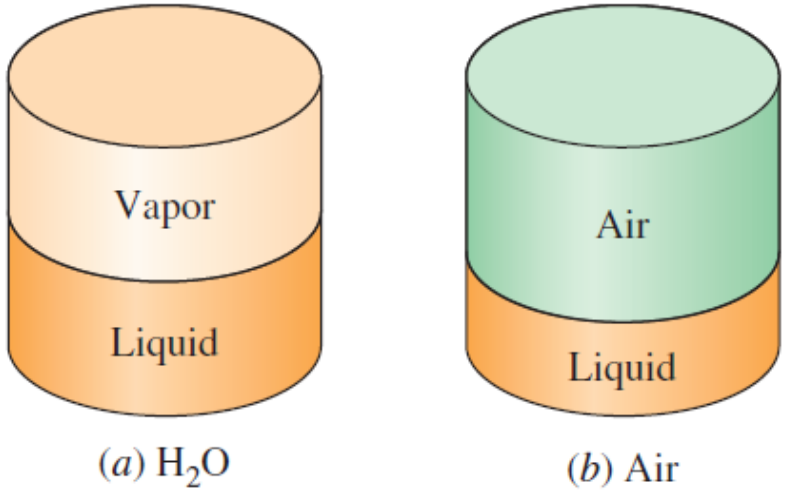


FIGURE 3–2
A mixture of liquid and gaseous water is a pure substance, but a mixture of liquid and gaseous air is not.

3.2 Phases of a Pure Substance

- A phase is identified as having a distinct molecular arrangement that is homogeneous throughout and separated from the others by easily identifiable boundary surfaces.
  - The two phases of H_2O in iced water represent a good example.
- Intermolecular bonds are strongest in solids and weakest in gases
- Gas phase are at a considerably higher energy level, thus gas must release a large amount of its energy before it can condense or freeze

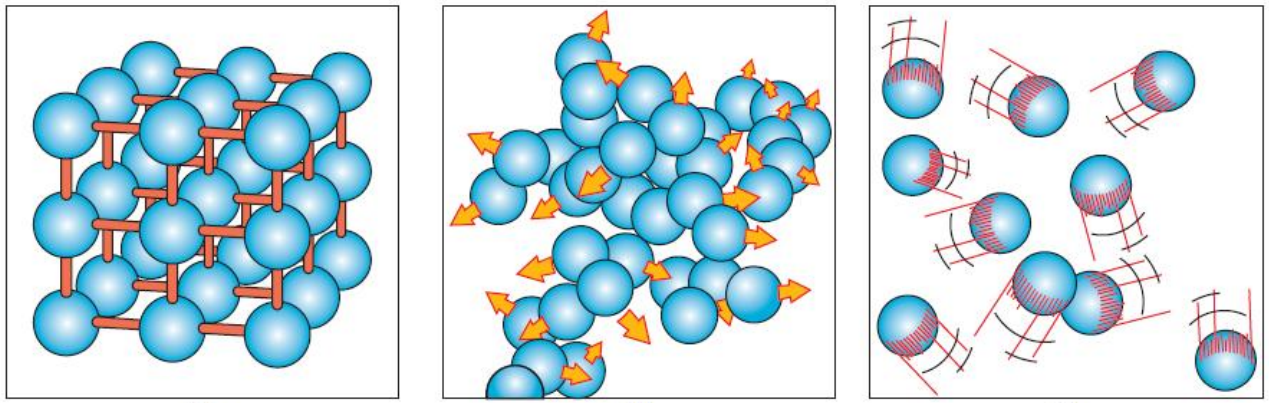
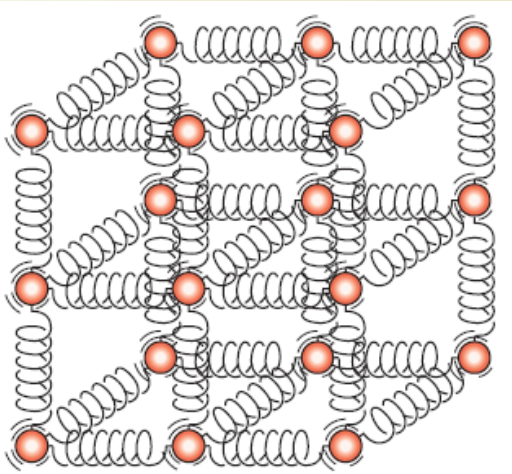


FIGURE 3–4
The arrangement of atoms in different phases: (*a*) molecules are at relatively fixed positions in a solid, (*b*) groups of molecules move about each other in the liquid phase, and (*c*) molecules move about at random in the gas phase.

The molecules in a solid are kept at their positions by the large springlike inter-molecular forces.



3.3 Phase-Change Processes of Pure Substances

- There are many practical situations where two phases of a pure substance coexist in equilibrium
- The process can be described by four stages
  1. Compressed liquid
  2. Saturated liquid (including saturated liquid–vapor mixture)
  3. Saturated vapor
  4. Superheated vapor

1. Consider a piston-cylinder device containing liquid water at 20°C and 1 atm pressure.
   - Under these conditions, water exists in the liquid phase, and it is called a **compressed liquid**, or a **subcooled liquid**, meaning that it is not about to vaporize

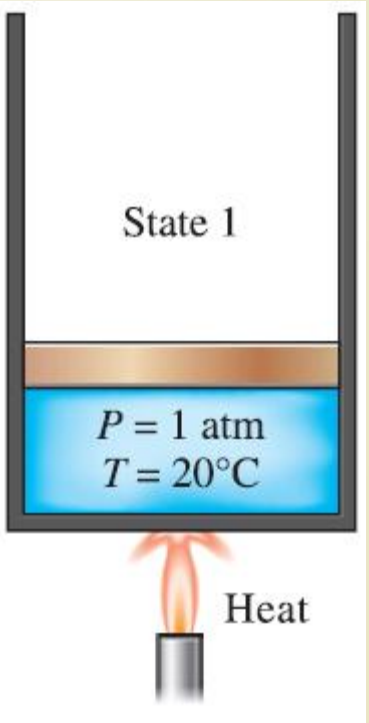


FIGURE 3–5
At 1 atm and 20°C, water exists in the liquid phase (*compressed liquid*).

3.3 Phase-Change Processes of Pure Substances (2)

2. Heat is now transferred to the water until it reaches 100°C.
   - At this state water is still a liquid, but any heat addition will cause some of the liquid to vaporize.
   - A liquid that is about to vaporize is called a **saturated liquid**

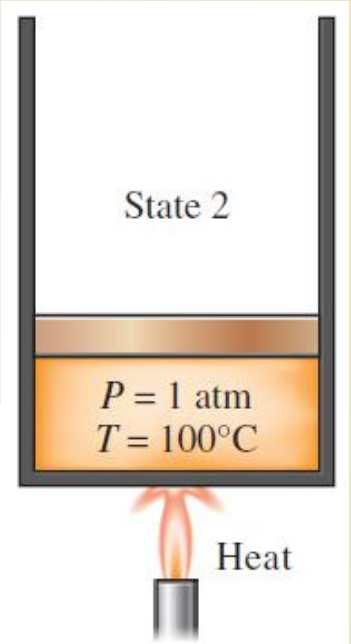


FIGURE 3–6
At 1 atm pressure and 100°C, water exists as a liquid that is ready to vaporize (*saturated liquid*).

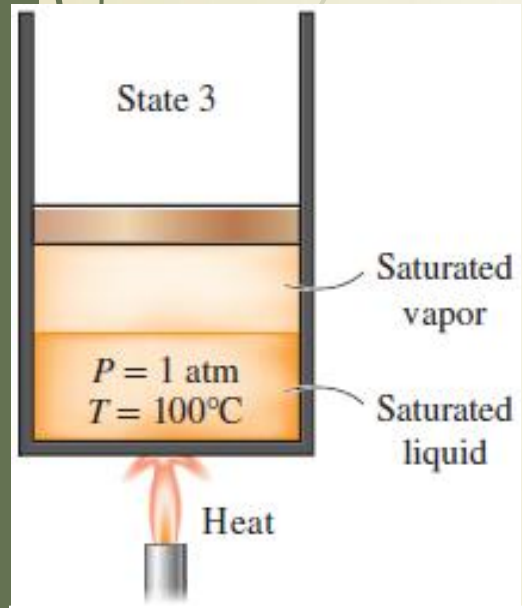


FIGURE 3–7
As more heat is transferred, part of the saturated liquid vaporizes (*saturated liquid–vapor mixture*).

3. Once boiling starts, the temperature stops rising (keeping pressure constant) until the liquid is completely vaporized.
   - During a boiling process, the only change we will observe is a large increase in the volume and a steady decline in the liquid level as a result of more liquid turning to vapor.
   - Midway about the vaporization line (state 3), the cylinder contains equal amounts of liquid and vapor.

3.3 Phase-Change Processes of Pure Substances (3)

4. As we continue transferring heat, the vaporization process continues until the last drop of liquid is vaporized.
   - At this point, the entire cylinder is filled with vapor that is on the borderline of the liquid phase.
   - Any **heat loss** from this vapor will cause some of the vapor to **condense**.
   - A **vapor that is about to condense** is called a **saturated vapor**.
   - A substance at **states between 2 and 4** is referred to as a **saturated liquid-vapor mixture** since the liquid and vapor phases coexist in equilibrium at these states.

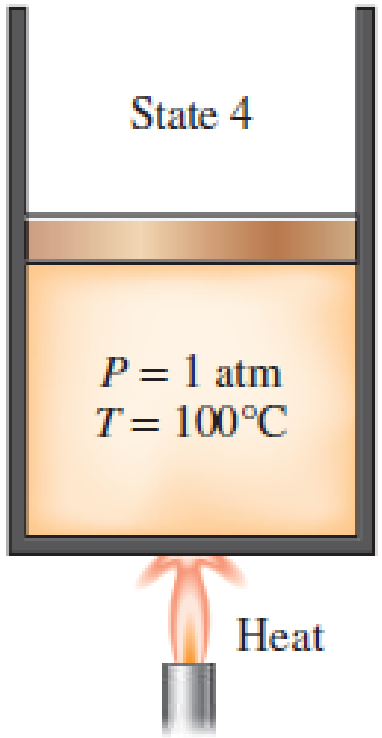


FIGURE 3–8
At 1 atm pressure, the temperature remains constant at 100°C until the last drop of liquid is vaporized (*saturated vapor*).

3.3 Phase-Change Processes of Pure Substances (3)

5. Once the phase-change process is completed, we are back to a single-phase region again (this time vapor)
   - Further transfer of heat results in an increase in both the temperature and the specific volume.
   - At state 5, if we transfer some heat from the vapor, the temperature may drop somewhat but condensation will not take place as long as the temperature remains above 100°C (for P = 1atm).
   - A vapor that is not about to condense (i.e., not a saturated vapor) is called a superheated vapor.
   - Therefore, water at state 5 is a **superheated vapor**.

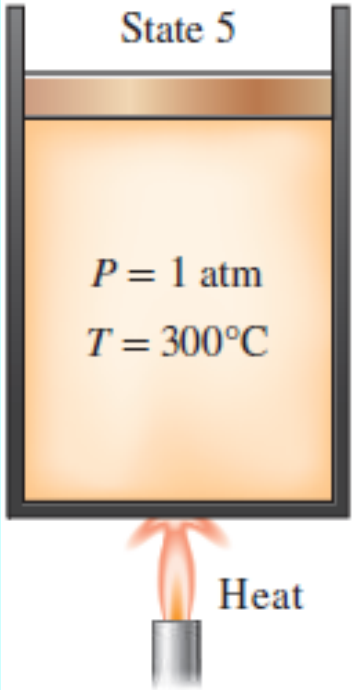


FIGURE 3–9
As more heat is transferred, the temperature of the vapor starts to rise (*superheated vapor*).

3.3 Phase-Change Processes of Pure Substances (4)

- The whole constant-pressure phase-change process can be illustrated on a T-V diagram
- If the entire process described here is reversed by cooling the water while maintaining the pressure at the same value,
  - the water will go back to state 1, retracing the same path, and
  - the amount of heat released will exactly match the amount of heat added during the heating process.

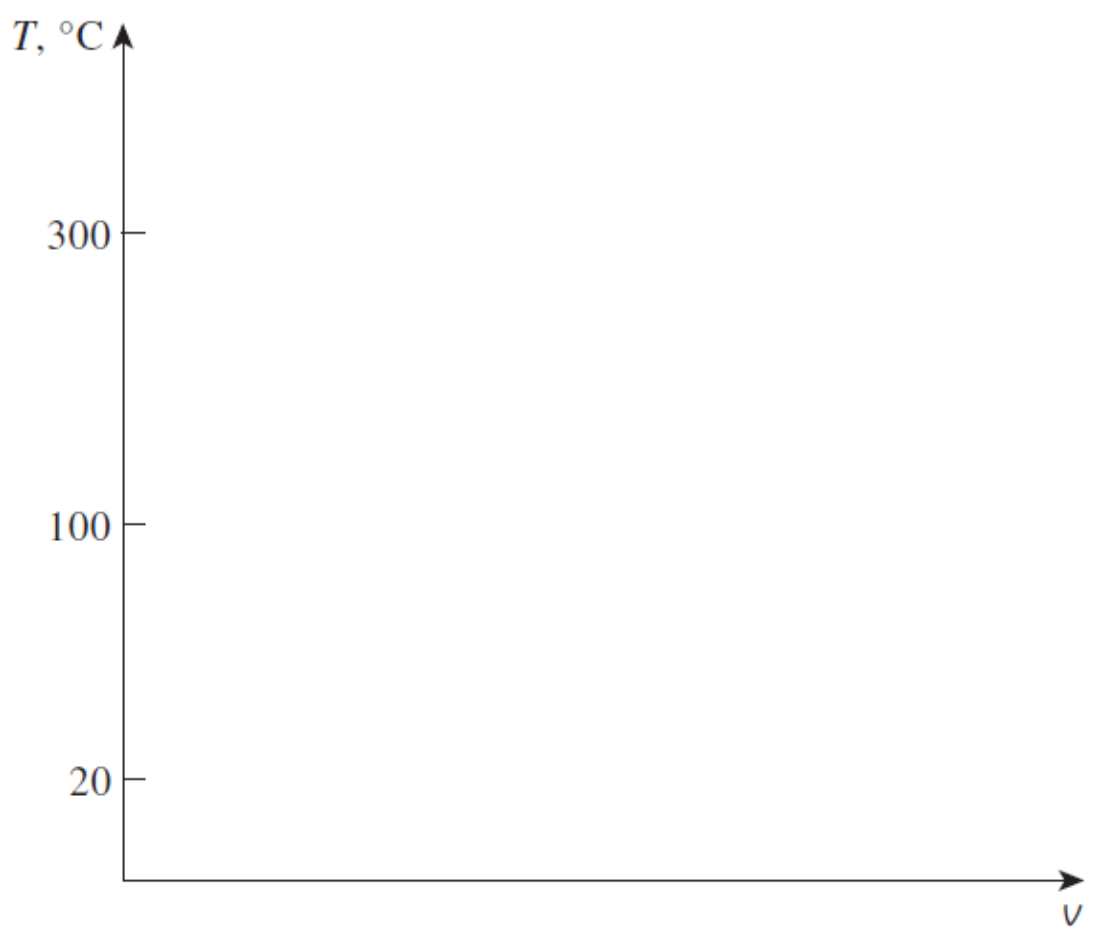


FIGURE 3–10
T-v diagram for the heating process of water at constant pressure.

3.3 Phase-Change Processes of Pure Substances (5)

Saturation Temperature and Saturation Pressure

- The temperature at which water starts boiling depends on the pressure; therefore, if pressure is fixed, so is the boiling temperature.
  - Water boils at 100°C at 1 atm pressure
  - At 500kPa, water boils at 151.8°C
- **Saturation temperature T_{sat}**: The temperature at which a pure substance changes phase at a given pressure.
- **Saturation pressure P_{sat}**: The pressure at which a pure substance changes phase at a given temperature.
- **Latent heat**: The amount of energy absorbed or released during a phase-change process.

TABLE 3–1

Saturation (or vapor) pressure of water at various temperatures

Temperature T, °C	Saturation Pressure P_{sat}, kPa
−10	0.260
−5	0.403
0	0.611
5	0.872
10	1.23
15	1.71
20	2.34
25	3.17
30	4.25
40	7.38
50	12.35
100	101.3 (1 atm)
150	475.8
200	1554
250	3973
300	8581

3.3 Phase-Change Processes of Pure Substances (6)

- **Latent heat of fusion**: The amount of energy absorbed during melting. It is equivalent to the amount of energy released during freezing.
- **Latent heat of vaporization**: The amount of energy absorbed during vaporization and it is equivalent to the energy released during condensation.
- The magnitudes of the latent heats depend on the temperature or pressure at which the phase change occurs.
- During a phase-change process, pressure and temperature are obviously dependent properties, and there is a definite relation between them, that is, $P_{sat} = f(T_{sat})$.
- A plot of P_{sat} versus T_{sat} is called a **liquid–vapor saturation curve**.

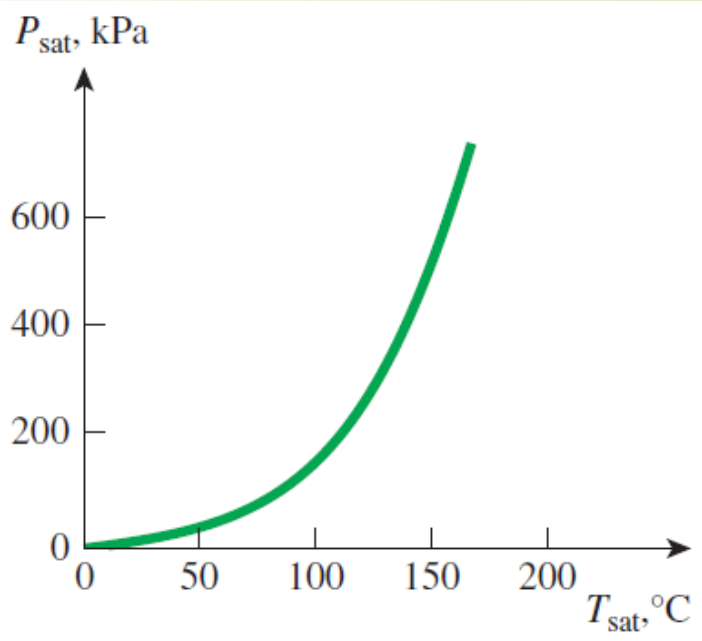


FIGURE 3–11
The liquid–vapor saturation curve of a pure substance (numerical values are for water).

3.3 Phase-Change Processes of Pure Substances (7)

Some Consequences of T_{sat} and P_{sat} Dependence

- As mentioned earlier, a substance at a specified pressure boils at the saturation temperature corresponding to that pressure.
- This phenomenon allows us to control the boiling temperature of a substance by simply controlling the pressure
- It has numerous applications in practice, and we shall discuss some of them

TABLE 3–2

Variation of the standard atmospheric pressure and the boiling (saturation) temperature of water with altitude

Elevation, m	Atmospheric pressure, kPa	Boiling temperature, °C
0	101.33	100.0
1,000	89.55	96.5
2,000	79.50	93.3
5,000	54.05	83.3
10,000	26.50	66.3
20,000	5.53	34.7

3.3 Phase-Change Processes of Pure Substances (8)

Case 1: A sealed can of liquid refrigerant-134a in a room at 25°C.

- If the can has been in the room long enough, the temperature of the refrigerant in the can is also 25°C.
- If the lid is opened slowly and some refrigerant is allowed to escape, the pressure in the can will start dropping until it reaches the atmospheric pressure.
  - Holding the can, you will notice its temperature dropping rapidly, and even ice forming outside the can if the air is humid.
  - Temperature in the can will register −26°C when the pressure drops to 1 atm, which is the saturation temperature of refrigerant-134a at that pressure.
  - The temperature of the liquid refrigerant will remain at −26°C until the last drop of it vaporizes.

TABLE A–11

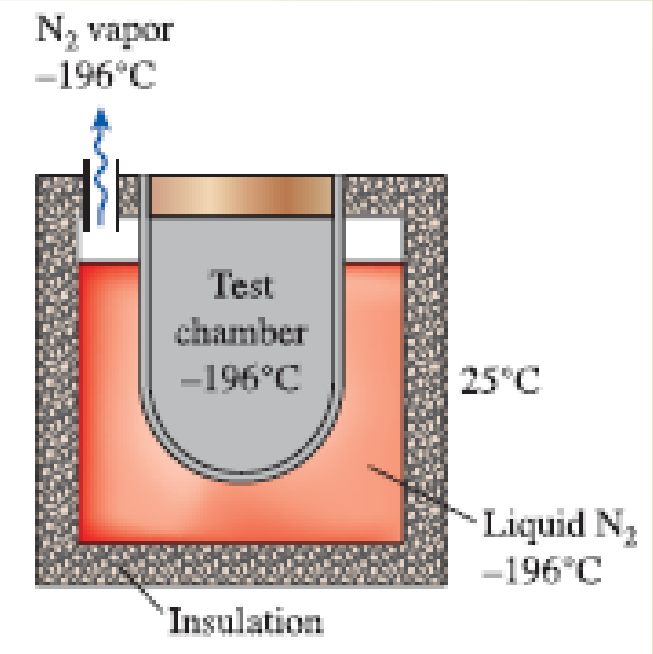
Saturated refrigerant-134a—Temperature table

		Specific volume, m³/kg		Inte
Temp., T °C	Sat. press., P_{sat} kPa	Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f
−40	51.25	0.0007054	0.36081	−0.036
−38	56.86	0.0007083	0.32732	2.475
−36	62.95	0.0007112	0.29751	4.992
−34	69.56	0.0007142	0.27090	7.517
−32	76.71	0.0007172	0.24711	10.05
−30	84.43	0.0007203	0.22580	12.59
−28	92.76	0.0007234	0.20666	15.13
−26	101.73	0.0007265	0.18946	17.69
−24	111.37	0.0007297	0.17395	20.25
−22	121.72	0.0007329	0.15995	22.82

3.3 Phase-Change Processes of Pure Substances (9)

Case 2: Liquid Nitrogen

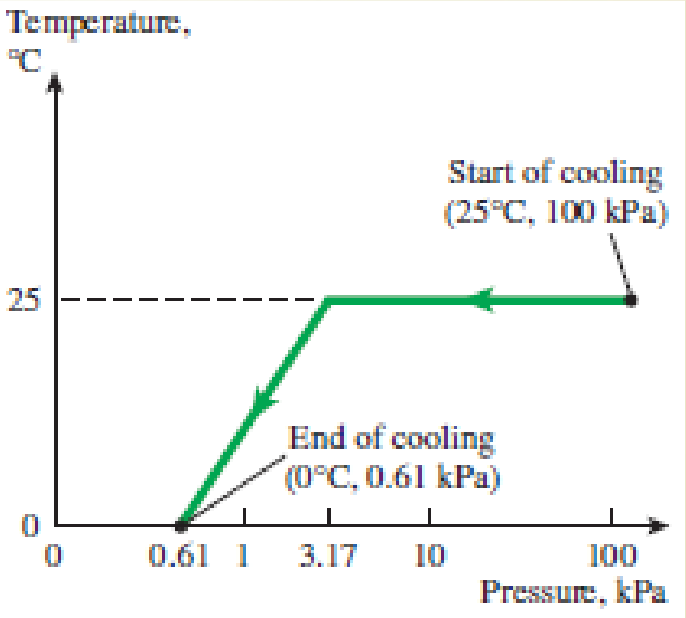
- The boiling temperature of nitrogen at atmospheric pressure is −196°C.
  - This means the temperature of liquid nitrogen exposed to the atmosphere must be −196°C since some nitrogen will be evaporating.
  - The temperature of liquid nitrogen remains constant at −196°C until it is depleted.
- For this reason, nitrogen is commonly used in low-temperature scientific studies (such as superconductivity) and cryogenic applications to maintain a test chamber at a constant temperature of −196°C.
- Liquid nitrogen is also used for medical purposes to burn off unsightly spots on the skin.
  - This is done by soaking a cotton swab in liquid nitrogen and wetting the target area with it.
  - As the nitrogen evaporates, it freezes the affected skin by rapidly absorbing heat from it.



3.3 Phase-Change Processes of Pure Substances (10)

Case 3: Vacuum Cooling

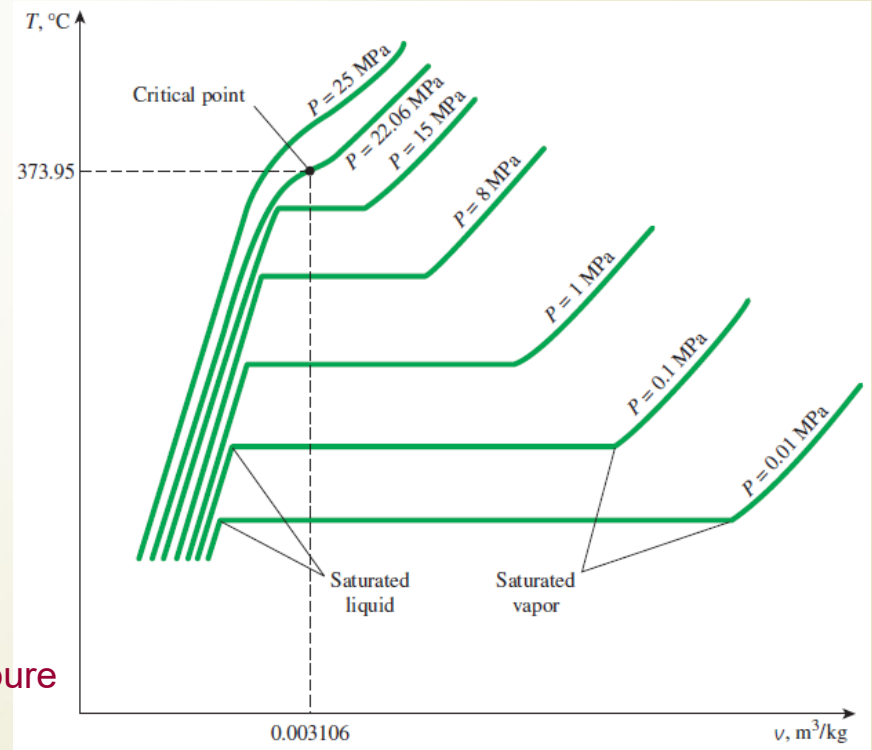
- A practical way of cooling leafy vegetables, based on reducing the pressure of the sealed cooling chamber to the saturation pressure at the desired low temperature and evaporating some water from the products to be cooled.
  - The heat of vaporization during evaporation is absorbed from the products, which lowers the product temperature.
  - The saturation pressure of water at 0°C is 0.61 kPa, and the products can be cooled to 0°C by lowering the pressure to this level.
- There are two distinct stages:
  - The products at ambient temperature, say at 25°C, are loaded into the chamber, and the operation begins. The temperature in the chamber remains constant until the saturation pressure is reached, which is 3.17 kPa at 25°C.
  - Saturation conditions are maintained inside at progressively lower pressures and temperatures until the desired temperature is reached.



3.4 Property Diagrams for Phase-change Processes

1. The *T-v* Diagram

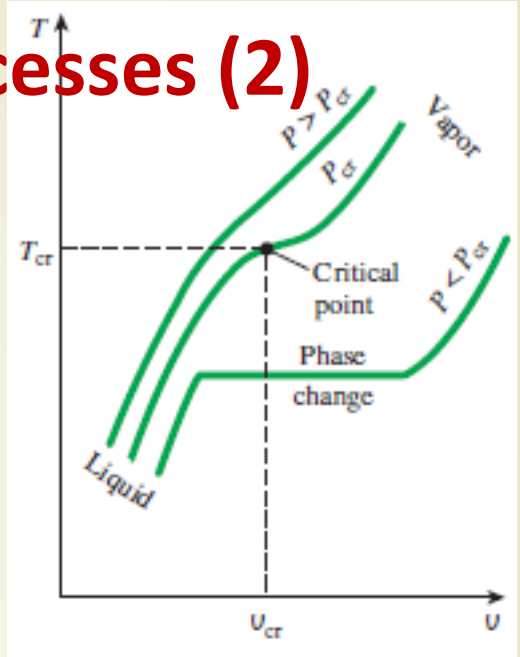
- As the pressure is increased, the saturation line (line joining saturated liquid and vapor states) continues to shrink and becomes a point when the pressure reaches 22.06 MPa for the case of water.
- This point is called the **critical point,** and it is defined as *the point at which the saturated liquid and saturated vapor states are identical.*
- The temperature, pressure, and specific volume of a substance at the critical point are:
  - the *critical temperature* T_{cr},
  - *critical pressure* P_{cr}, and
  - *critical specific volume* v_{cr}.
  - For water P_{cr} = 22.06 MPa, T_{cr} = 373.95°C, and v_{cr} = 0.003106 m3/kg.
- The critical properties for various substances are given in Appendix (Table A–1)

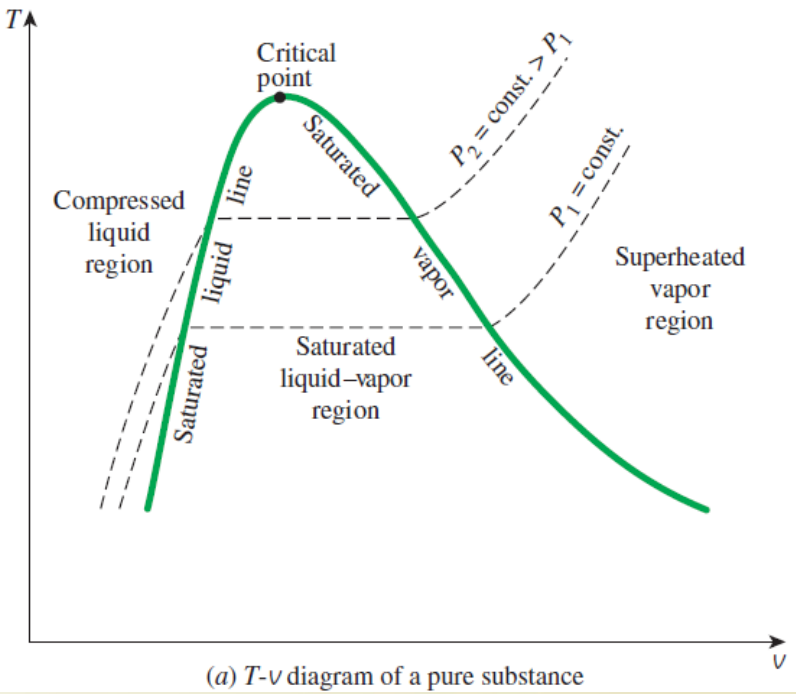


T-v diagram of constant-pressure phase-change processes of a pure substance at various pressures (numerical values are for water).

3.4 Property Diagrams for Phase-change Processes (2)

- Above the critical state, there is no line that separates the compressed liquid region and the superheated vapor region.
- However, it is customary to refer to the substance as superheated vapor at temperatures above the critical temp. and as compressed liquid at temps. below the critical temp.
- The saturated liquid states can be connected by a line called the **saturated liquid line,** and the saturated vapor states in the same figure can be connected by another line, called the **saturated vapor line.**
- These two lines meet at the critical point, forming a dome, called **the saturated liquid–vapor mixture region**, or **the wet region**.



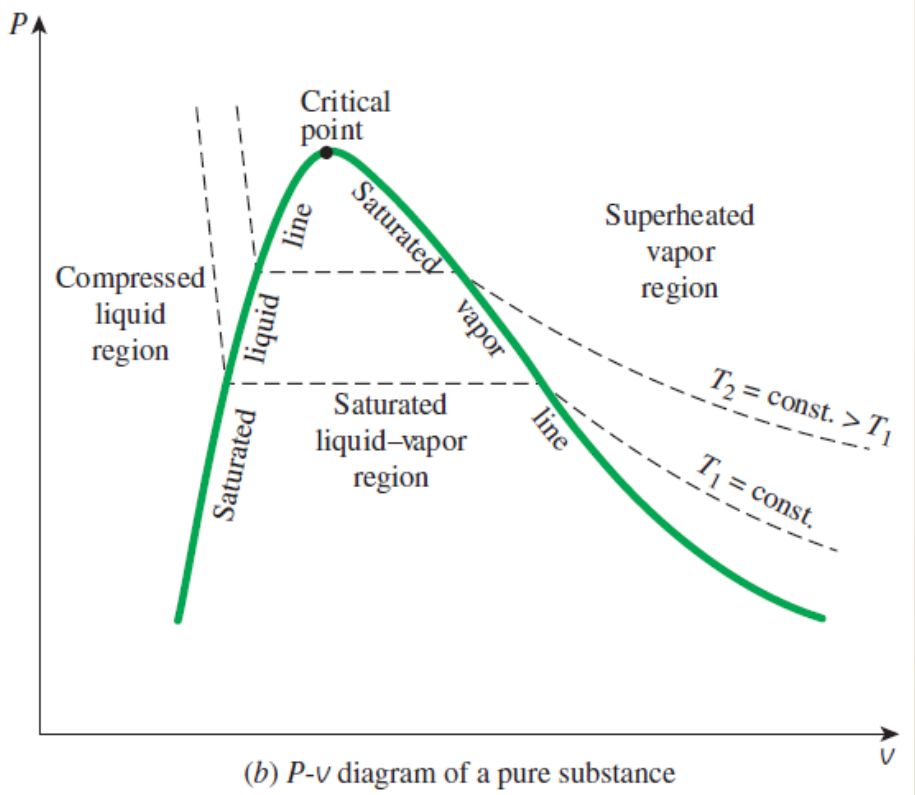


(a) T-v diagram of a pure substance

3.4 Property Diagrams for Phase-change Processes (3)

2. The *P-v* Diagram

- The general shape of the *P-v* diagram of a pure substance is very much like the *T-v* diagram, but the *T* = constant lines on this diagram have a downward trend.

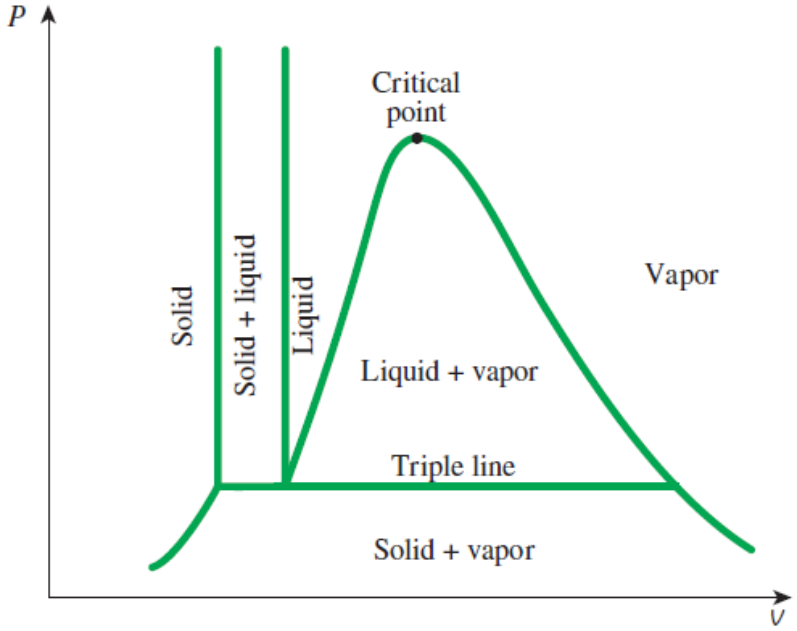


(*b*) *P-v* diagram of a pure substance

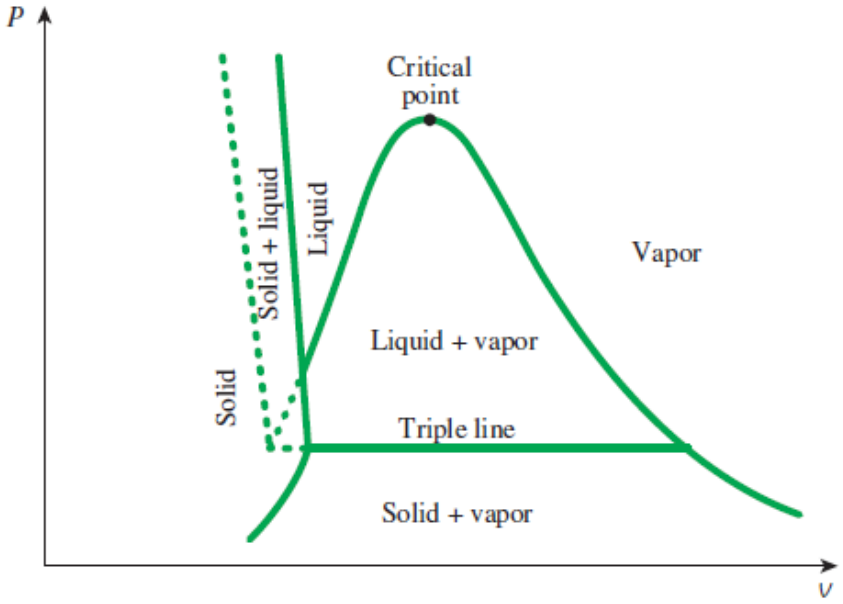
3.4 Property Diagrams for Phase-change Processes (4)

Extending the Diagrams to Include the Solid Phase

- Most substances contract during a solidification (i.e., freezing) process. Others, like water, expand as they freeze.
- The *P–v* diagrams for both groups of substances are given below.
- These two diagrams differ only in the solid-liquid saturation region. The *T-v* diagrams look very much like the *P-v* diagrams.



(*a*) *P-v* diagram of a substance that contracts on freezing



(*b*) *P-v* diagram of a substance that expands on freezing (such as water)

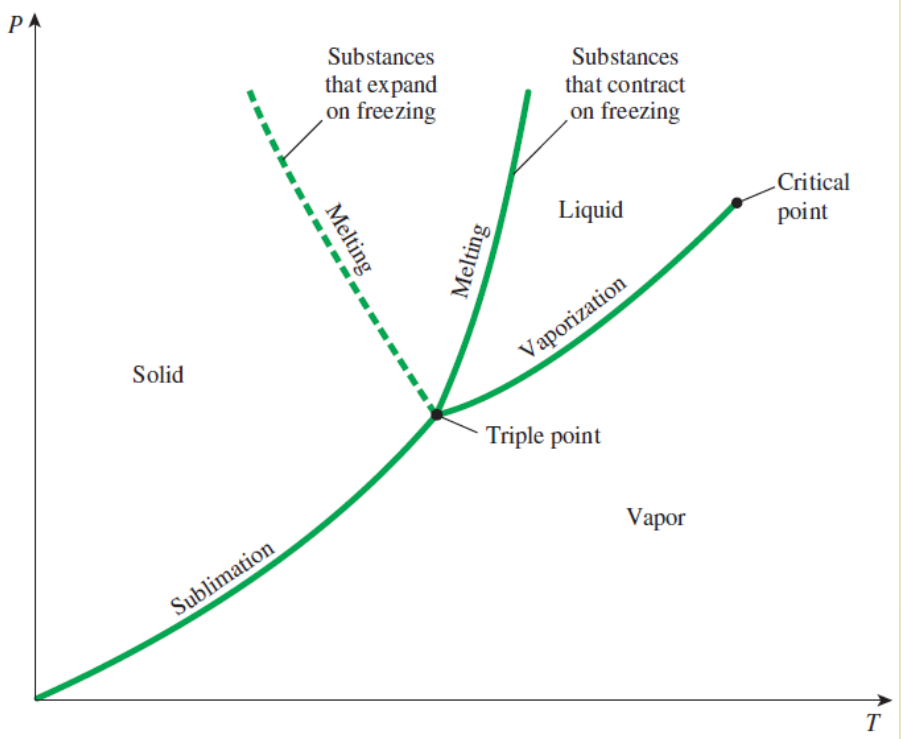
3.4 Property Diagrams for Phase-change Processes (5)

- Under some conditions all three phases of a pure substance coexist in equilibrium.
- On *P-v* or *T-v* diagrams, these triple-phase states form a line called the triple line.
- The states on the triple line of a substance have the same pressure and temperature but different specific volumes.
- The triple line appears as a point on the *P-T* diagrams and, therefore, is often called the triple point.
- **Sublimation** is the process of passing from the solid phase directly into the vapor phase (example: solid CO_2 (dry ice)).
    - occurs at pressures below the triple point value, since a pure substance cannot exist in the liquid phase at those pressures
- *Sublimation line* separates the solid and vapor regions on the phase diagram.

3.4 Property Diagrams for Phase-change Processes (6)

3. The *P-T* Diagram

- This diagram is often called the phase diagram since all three phases are separated from each other by three lines.
- The sublimation line separates the solid and vapor regions, the vaporization line separates the liquid and vapor regions, and the melting (or fusion) line separates the solid and liquid regions.
- These three lines meet at the triple point, where all three phases coexist in equilibrium.
- Substances that expand and contract on freezing differ only in the melting line on the *P-T* diagram.



P-T diagram of pure substances.

3.4 Property Diagrams for Phase-change Processes (7)

The P-v-T Surface

- All the two-dimensional diagrams we have discussed so far are merely projections of this 3-dimensional surface onto the appropriate planes.
- All the points on the surface represent equilibrium states.
- The *P-v-T* surfaces present a great deal of information at once, but in a thermodynamic analysis it is more convenient to work with two-dimensional diagrams, such as *P-v* and *T-v* diagrams.

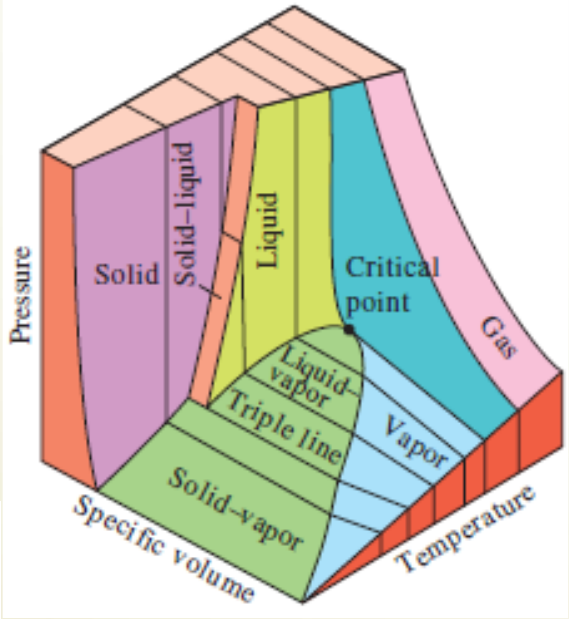


FIGURE 3–23
P-v-T surface of a substance that *contracts* on freezing.

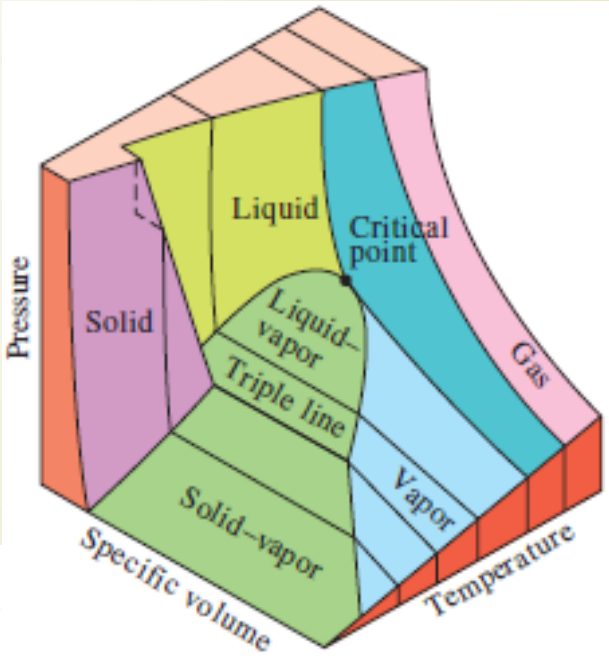


FIGURE 3–24
P-v-T surface of a substance that expands on freezing (like water).

3.5 Property Tables

- For most substances, the relationships among thermodynamic properties are too complex to be expressed by simple equations.
- Therefore, properties are frequently presented in the form of tables.
- In the following discussion, the steam tables are used to demonstrate the use of thermodynamic property tables.
- Property tables of other substances are used in the same manner.
- For each substance, the thermodynamic properties are listed in a separate table for each region of interest such as the superheated vapor, compressed liquid, and saturated (mixture) regions.
- Before we get into the discussion of property tables, we define a new property called ***enthalpy***.

3.5 Property Tables (2)

Enthalpy-A Combination property

- In the analysis of certain types of processes, particularly in power generation and refrigeration, we often encounter the combination of properties $u + Pv$.
- This combination is defined as a new property, **enthalpy**, and given the symbol h:

$$h = u + Pv \quad \text{(kJ/kg)} \qquad \text{- Specific enthalpy}$$

or,

$$H = U + PV \quad \text{(kJ)} \qquad \text{- Total enthalpy}$$

- It was introduced by Prof. Richard Mollier, who recognized the importance of the group $u + Pv$ in the analysis of steam turbines.

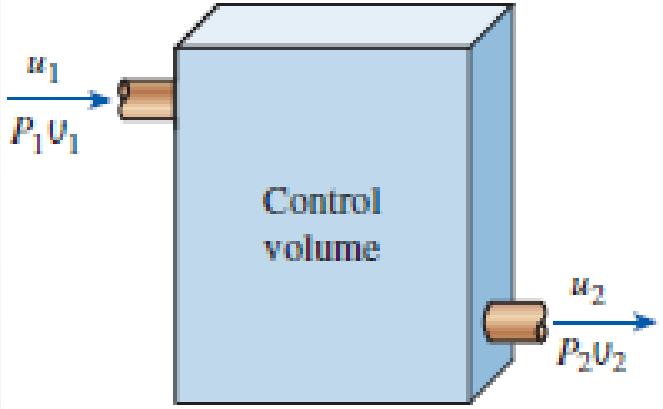


FIGURE 3–25
The combination $u + Pv$ is often encountered in the analysis of control volumes.

3.5 Property Tables (3)

1a Saturated Liquid and Saturated Vapor States

- The properties of saturated liquid and saturated vapor for water are listed in Tables A-4 (under *T*) and A-5 (under *P*).
  - v_f = specific volume of saturated liquid
  - v_g = specific volume of saturated vapor
  - v_{fg} = difference between vg and vf (i.e., $v_{fg} = v_g - v_f$)
- The quantity h_{fg} is called the enthalpy of vaporization (or latent heat of vaporization).
  - It represents the amount of energy needed to vaporize a unit mass of saturated liquid at a given temp. or pressure.
  - It decreases as the temp. or pressure increases and becomes zero at the critical point.

Temp. °C T	Sat. press. kPa P_{sat}	Specific volume m³/kg Sat. liquid v_f	Specific volume m³/kg Sat. vapor v_g
85	57.868	0.001032	2.8261
90	70.183	0.001036	2.3593
95	84.609	0.001040	1.9808

Specific temperature

Corresponding saturation pressure

Specific volume of saturated liquid

Specific volume of saturated vapor

A partial list of Table A-4.

3.5 Property Tables (4)

1b Saturated Liquid-Vapor Mixture

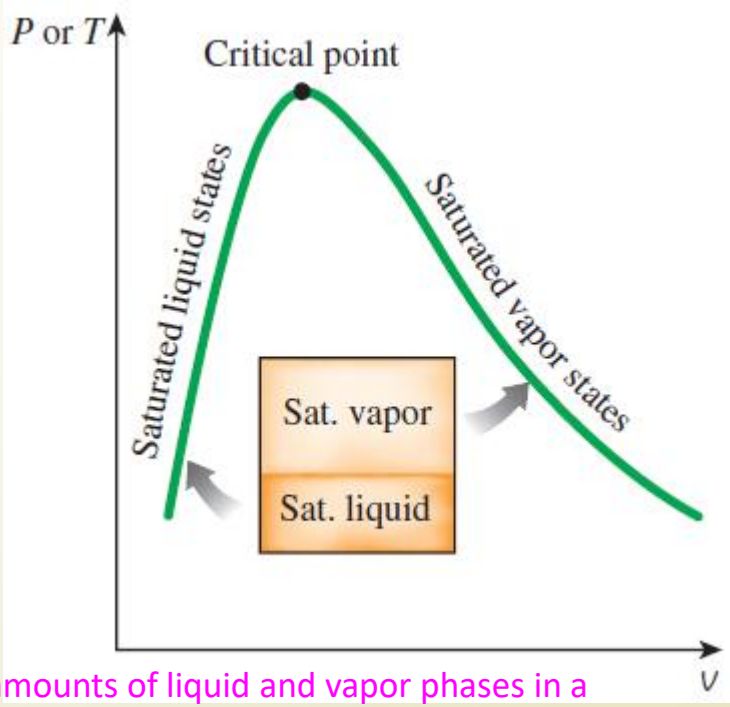
- During a vaporization process, a substance exists as part liquid and part vapor.
- To analyze this mixture properly, we need to know the proportions of the liquid and vapor phases in the mixture.
- This is done by defining a new property called the **quality *x*** as the ratio of the mass of vapor to the total mass of the mixture:

$$x = \frac{m_{vapor}}{m_{total}} = \frac{m_g}{m_t}$$

where

$$m_{\text{total}} = m_{\text{liquid}} + m_{\text{vapor}} = m_f + m_g$$

- The quality, *x* is between 0 and 1;
  - 0 – saturated liquid.
  - 1 – saturated vapor.



The relative amounts of liquid and vapor phases in a saturated mixture are specified by the *quality x*.

3.5 Property Tables (5)

Consider a tank that contains a saturated liquid–vapor mixture. The volume occupied by saturated liquid is V_f, and the volume occupied by saturated vapor is V_g. The total volume V is the sum of the two:

A two phase system can be treated as a homogeneous mixture

$$V = V_f + V_g$$

$$V = mv \longrightarrow m_t v_{avg} = m_f v_f + m_g v_g$$

$$m_f = m_t - m_g \longrightarrow m_t v_{avg} = (m_t - m_g) v_f + m_g v_g$$

Dividing by m_t yields

$$v_{avg} = (1 - x) v_f + x v_g = v_f + x(v_g - v_f)$$

since $x = m_g/m_t$. This relation can also be expressed as

$$v_{avg} = v_f + x v_{fg} \qquad (\mathrm{m^3/kg}) \qquad (3\text{–}4)$$

where $v_{fg} = v_g - v_f$. Solving for quality, we obtain

$$x = \frac{v_{avg} - v_f}{v_{fg}} \qquad (3\text{–}5)$$

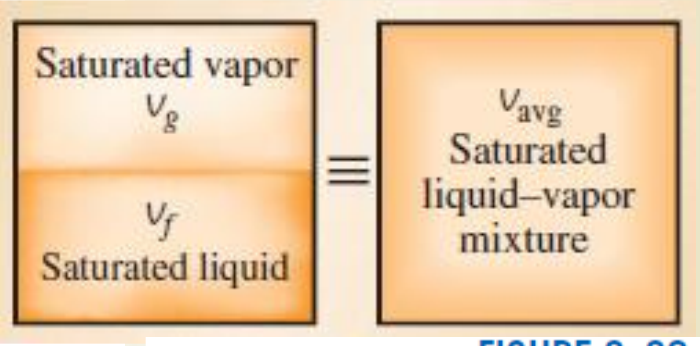


FIGURE 3–32
A two-phase system can be treated as a homogeneous mixture for convenience.

3.5 Property Tables (6)

- Based on this equation, quality can be related to the horizontal distances on a *P-v* or *T-v* diagram (Fig. 3–33).
- The analysis can be repeated for internal energy and enthalpy with the following results:

$$u_{avg} = u_f + xu_{fg} \quad \text{(kJ/kg)}$$

$$h_{avg} = h_f + xh_{fg} \quad \text{(kJ/kg)}$$

- Or can be summarized as:

$y \rightarrow v$, u, or h.

$$y_{avg} = y_f + xy_{fg}$$

$$y_f \leq y_{avg} \leq y_g$$

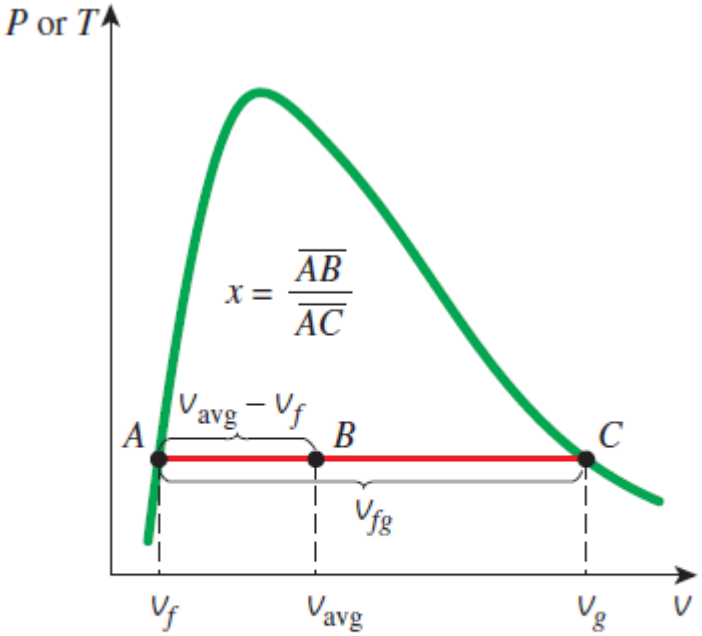


FIGURE 3–33
Quality is related to the horizontal distances on *P-v* and *T-v* diagrams.

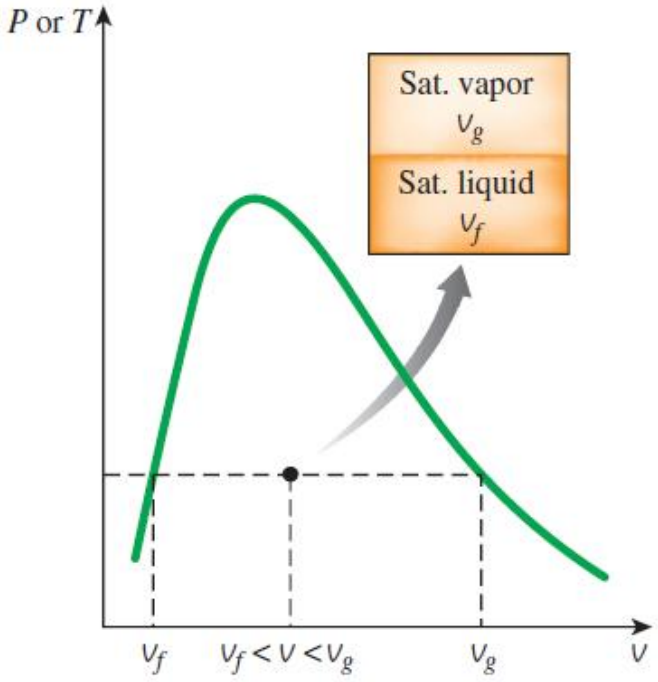


FIGURE 3–34
The *v* value of a saturated liquid–vapor mixture lies between the v_f and v_g values at the specified *T* or *P*.

EXAMPLE 3–4 Pressure and Volume of a Saturated Mixture

A rigid tank contains 10 kg of water at 90°C. If 8 kg of the water is in the liquid form and the rest is in the vapor form, determine (*a*) the pressure in the tank and (*b*) the volume of the tank.

SOLUTION A rigid tank contains saturated mixture. The pressure and the volume of the tank are to be determined.

Analysis (*a*) The state of the saturated liquid–vapor mixture is shown in Fig. 3–35. Since the two phases coexist in equilibrium, we have a saturated mixture, and the pressure must be the saturation pressure at the given temperature:

$$P = P_{\text{sat @ 90°C}} = \mathbf{70.183\ kPa} \qquad \text{(Table A–4)}$$

(*b*) At 90°C, we have $v_f = 0.001036\ \text{m}^3/\text{kg}$ and $v_g = 2.3593\ \text{m}^3/\text{kg}$ (Table A–4). One way of finding the volume of the tank is to determine the volume occupied by each phase and then add them:

$$\begin{aligned} V &= V_f + V_g = m_f v_f + m_g v_g \\ &= (8\ \text{kg})(0.001036\ \text{m}^3/\text{kg}) + (2\ \text{kg})(2.3593\ \text{m}^3/\text{kg}) \\ &= \mathbf{4.73\ m^3} \end{aligned}$$

Another way is to first determine the quality x, then the average specific volume v, and finally the total volume:

$$x = \frac{m_g}{m_t} = \frac{2\ \text{kg}}{10\ \text{kg}} = 0.2$$

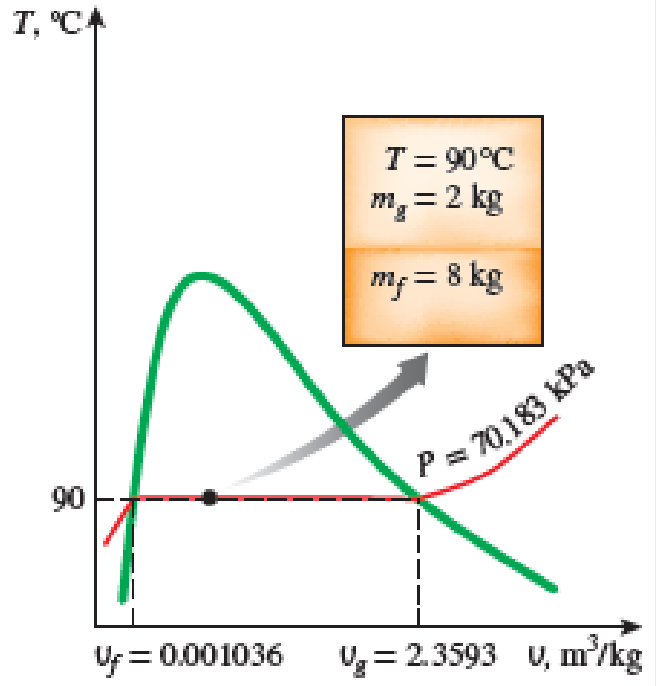


FIGURE 3–35
Schematic and *T*-*v* diagram for Example 3–4.

$$v = v_f + x v_{fg}$$
$$= 0.001036 \text{ m}^3/\text{kg} + (0.2)[(2.3593 - 0.001036) \text{ m}^3/\text{kg}]$$
$$= 0.473 \text{ m}^3/\text{kg}$$

and

$$V = mv = (10 \text{ kg})(0.473 \text{ m}^3/\text{kg}) = 4.73 \text{ m}^3$$

Discussion The first method appears to be easier in this case since the masses of each phase are given. In most cases, however, the masses of each phase are not available, and the second method becomes more convenient.

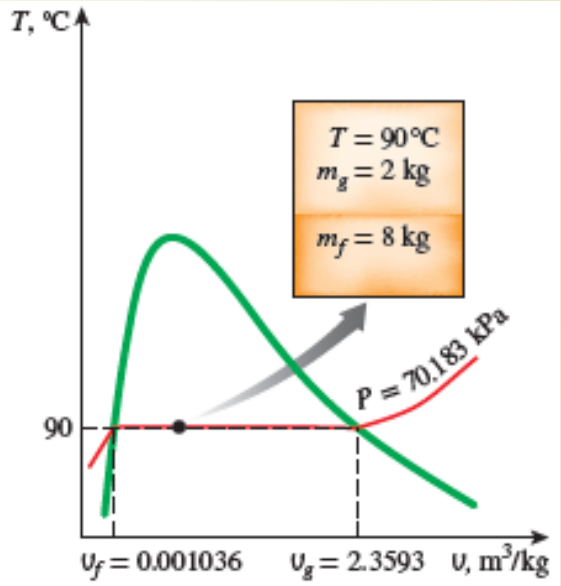


FIGURE 3–35
Schematic and T-v diagram for Example 3–4.

EXAMPLE 3–5 Properties of Saturated Liquid–Vapor Mixture

An 80-L vessel contains 4 kg of refrigerant-134a at a pressure of 160 kPa. Determine (*a*) the temperature, (*b*) the quality, (*c*) the enthalpy of the refrigerant, and (*d*) the volume occupied by the vapor phase.

SOLUTION A vessel is filled with refrigerant-134a. Some properties of the refrigerant are to be determined.

Analysis (*a*) The state of the saturated liquid–vapor mixture is shown in Fig. 3–36. At this point we do not know whether the refrigerant is in the compressed liquid, superheated vapor, or saturated mixture region. This can be determined by comparing a suitable property to the saturated liquid and saturated vapor values. From the information given, we can determine the specific volume:

$$v = \frac{V}{m} = \frac{0.080\ \text{m}^3}{4\ \text{kg}} = 0.02\ \text{m}^3/\text{kg}$$

At 160 kPa, we read

$$v_f = 0.0007435\ \text{m}^3/\text{kg}$$
$$v_g = 0.12355\ \text{m}^3/\text{kg} \qquad \text{(Table A–12)}$$

Obviously, $v_f < v < v_g$, and the refrigerant is in the saturated mixture region. Thus, the temperature must be the saturation temperature at the specified pressure:

$$T = T_{\text{sat @ 160 kPa}} = -15.60°\text{C}$$

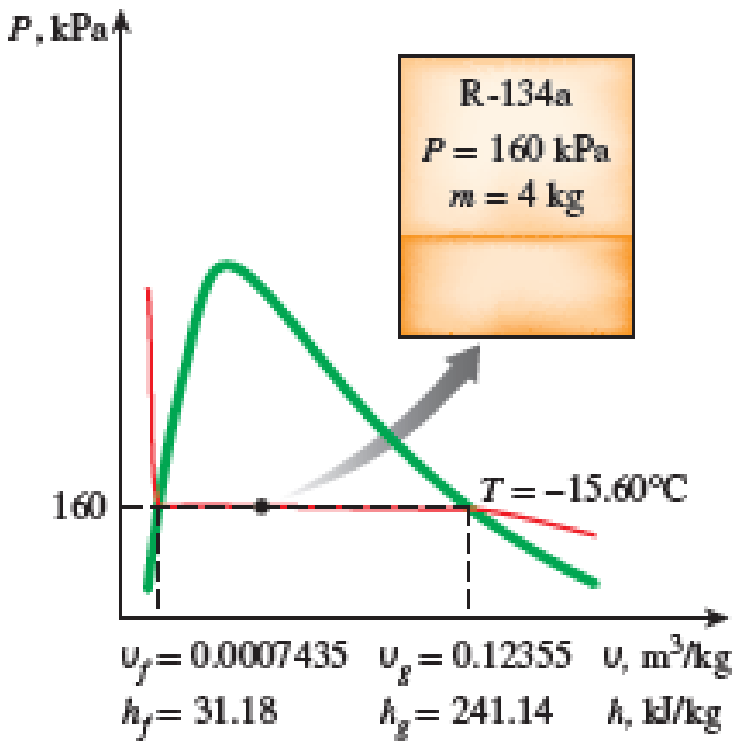


FIGURE 3–36
Schematic and *P*-*v* diagram for Example 3–5.

(*b*) Quality can be determined from

$$x = \frac{v - v_f}{v_{fg}} = \frac{0.02 - 0.0007435}{0.12355 - 0.0007435} = \mathbf{0.157}$$

(*c*) At 160 kPa, we also read from Table A–12 that $h_f = 31.18$ kJ/kg and $h_{fg} = 209.96$ kJ/kg. Then,

$$\begin{aligned} h &= h_f + x h_{fg} \\ &= 31.18 \text{ kJ/kg} + (0.157)(209.96 \text{ kJ/kg}) \\ &= \mathbf{64.1 \text{ kJ/kg}} \end{aligned}$$

(*d*) The mass of the vapor is

$$m_g = x m_t = (0.157)(4 \text{ kg}) = 0.628 \text{ kg}$$

and the volume occupied by the vapor phase is

$$V_g = m_g v_g = (0.628 \text{ kg})(0.12355 \text{ m}^3/\text{kg}) = \mathbf{0.0776 \text{ m}^3} \text{ (or 77.6 L)}$$

The rest of the volume (2.4 L) is occupied by the liquid.

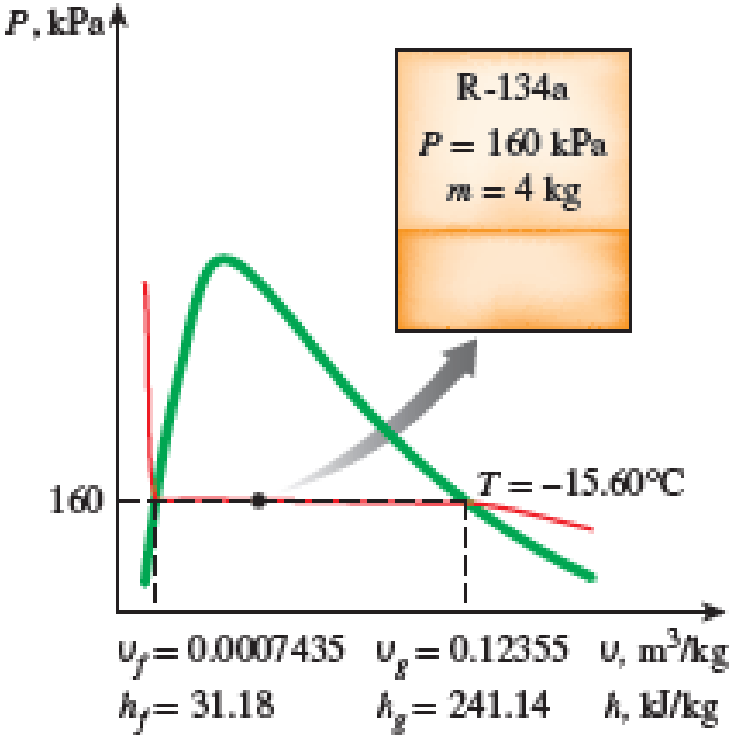
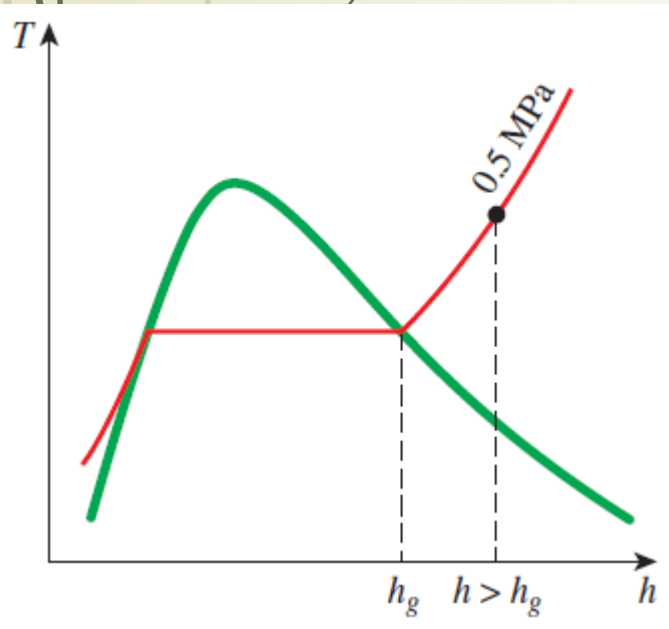


FIGURE 3–36
Schematic and *P*-*v* diagram for Example 3–5.

3.5 Property Tables (7)

2 Superheated Vapor

- In the region to the right of the saturated vapor line and at temperatures above the critical point temperature, a substance exists as **superheated vapor**.
- In this region, temperature and pressure are independent properties.
- Compared to saturated vapor, superheated vapor is characterized by:



Lower pressures ($P < P_{sat}$ at a given T)
Higher tempreatures ($T > T_{sat}$ at a given P)
Higher specific volumes ($v > v_g$ at a given P or T)
Higher internal energies ($u > u_g$ at a given P or T)
Higher enthalpies ($h > h_g$ at a given P or T)

T,°C	v m³/kg	u kJ/kg	h kJ/kg
	P = 0.1 MPa (99.61°C)		
Sat.	1.6941	2505.6	2675.0
100	1.6959	2506.2	2675.8
150	1.9367	2582.9	2776.6
⋮	⋮	⋮	⋮
1300	7.2605	4687.2	5413.3
	P = 0.5 MPa (151.83°C)		
Sat.	0.37483	2560.7	2748.1
200	0.42503	2643.3	2855.8
250	0.47443	2723.8	2961.0

A partial listing of Table A–6.

EXAMPLE 3–7 Temperature of Superheated Vapor

Determine the temperature of water at a state of $P = 0.5$ MPa and $h = 2890$ kJ/kg.

SOLUTION The temperature of water at a specified state is to be determined.

Analysis At 0.5 MPa, the enthalpy of saturated water vapor is $h_g = 2748.1$ kJ/kg. Since $h > h_g$, as shown in Fig. 3–39, we again have superheated vapor. Under 0.5 MPa in Table A–6 we read

T, °C	h, kJ/kg
200	2855.8
250	2961.0

Obviously, the temperature is between 200 and 250°C. By linear interpolation it is determined to be

$$T = \mathbf{216.3°C}$$

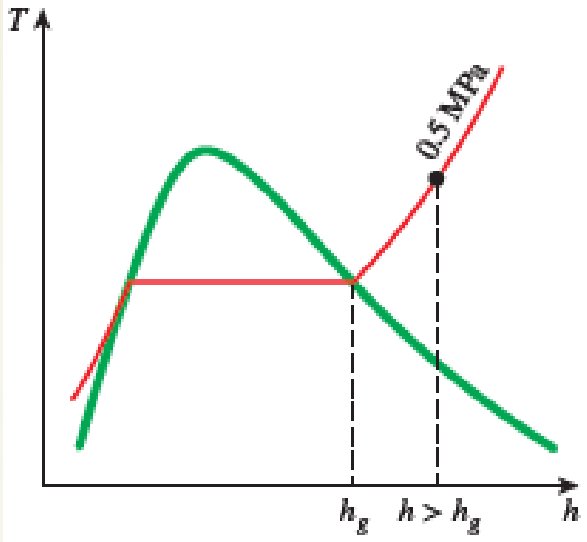


FIGURE 3–39
At a specified P, superheated vapor exists at a higher h than the saturated vapor (Example 3–7).

3.5 Property Tables (8)

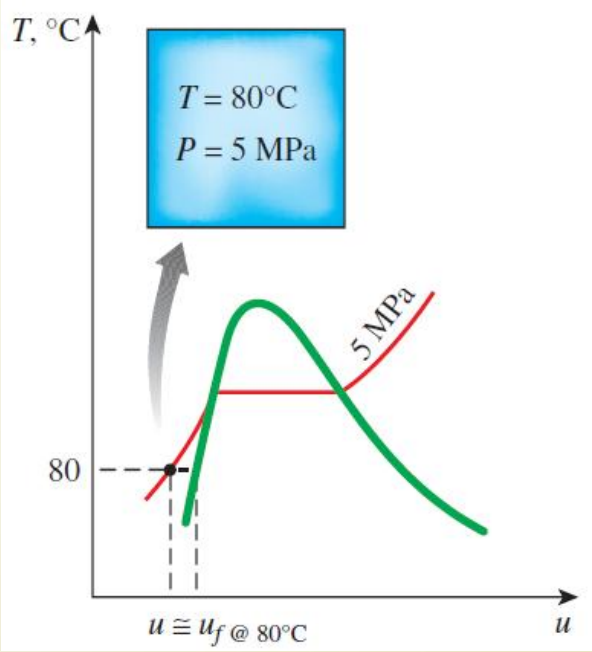
3 Compressed Liquid

- Compressed liquid tables are not as commonly available
- Variation of properties of compressed liquid with pressure is very mild.
  - Increasing the pressure 100 times often causes properties to change less than 1%.
- The properties depend on temperature much more strongly than pressure.

$$y \cong y_{f\,@\,T} \qquad y \rightarrow v,\ u,\ \text{or}\ h$$

- For low to moderate pressures and temperatures, *h* is more accurately evaluated using: $h \cong h_{f\,@\,T} + v_{f\,@\,T}(P - P_{\text{sat}\,@T})$
- Compressed liquid is characterized by:

Higher pressures ($P > P_{\text{sat}}$ at a given T)
Lower tempreatures ($T < T_{\text{sat}}$ at a given P)
Lower specific volumes ($v < v_f$ at a given P or T)
Lower internal energies ($u < u_f$ at a given P or T)
Lower enthalpies ($h < h_f$ at a given P or T)



EXAMPLE 3–8 Approximating Compressed Liquid as Saturated Liquid

Determine the internal energy of compressed liquid water at 80°C and 5 MPa, using (*a*) data from the compressed liquid table and (*b*) saturated liquid data. What is the error involved in the second case?

SOLUTION The exact and approximate values of the internal energy of liquid water are to be determined.

Analysis At 80°C, the saturation pressure of water is 47.416 kPa, and since 5 MPa > P_{sat}, we obviously have compressed liquid, as shown in Fig. 3–41.

(*a*) From the compressed liquid table (Table A–7)

$$\left.\begin{array}{l} P = 5 \text{ MPa} \\ T = 80°\text{C} \end{array}\right\} \quad u = \mathbf{333.82\ kJ/kg}$$

(*b*) From the saturation table (Table A–4), we read

$$u \cong u_{f\,@\,80°\text{C}} = \mathbf{334.97\ kJ/kg}$$

The error involved is

$$\frac{334.97 - 333.82}{333.82} \times 100 = \mathbf{0.34\%}$$

which is less than 1 percent.

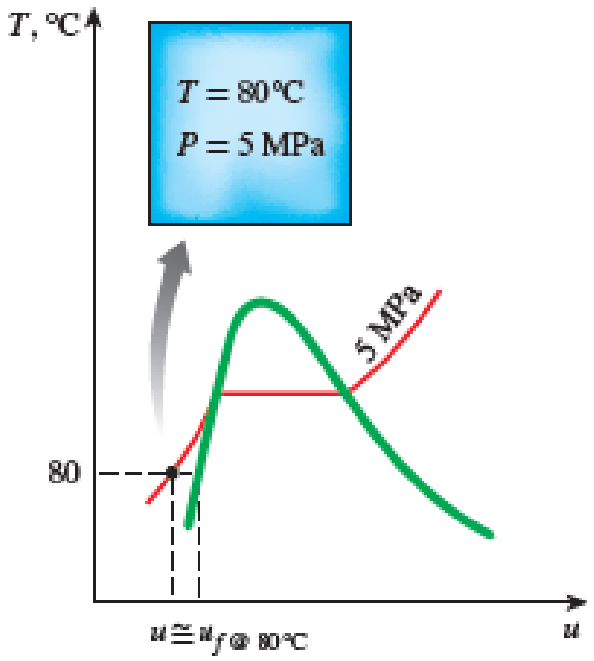


FIGURE 3–41
Schematic and *T*-*u* diagram for Example 3–8.

3.5 Property Tables (8)

Reference State and Reference Values

- The values of *u*, *h*, and *s* cannot be measured directly, and they are calculated from measurable properties using the relations between properties.
- However, those relations give the changes in properties, not the values of properties at specified states.
- Therefore, a convenient reference state need to be assigned a value of zero as reference.
    - The reference state for water is 0.01°C and for R-134a is -40°C in tables.
    - Some properties may have negative values due to the chosen reference state.
    - Different tables (textbooks) may use a different reference state, thus will list different values for some properties.
    - However, In thermodynamics we are concerned with the changes in properties, and the reference state chosen is of no consequence in calculations.

Saturated water—Temperature table

		Specific volume, m^3/kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg · K		
Temp., T °C	Sat. press., P_{sat} kPa	Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
0.01	0.6117	0.001000	206.00	0.000	2374.9	2374.9	0.001	2500.9	2500.9	0.0000	9.1556	9.1556
5	0.8725	0.001000	147.03	21.019	2360.8	2381.8	21.020	2489.1	2510.1	0.0763	8.9487	9.0249

Saturated refrigerant-134a—Temperature table

		Specific volume, m^3/kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg · K		
Temp., T °C	Sat. press., P_{sat} kPa	Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
−40	51.25	0.0007054	0.36081	−0.036	207.40	207.37	0.000	225.86	225.86	0.00000	0.96866	0.96866

Exercise 1

3–22 Complete this table for H_2O:

T, °C	P, kPa	v, m^3/kg	Phase description
140		0.035	
	550		Saturated liquid
125	750		
300		0.140	

Exercise 2

3–25 Complete this table for H_2O:

T, °C	P, kPa	h, kJ/kg	x	Phase description
	200		0.7	
140		1800		
	950		0.0	
80	500			
	800	3162.2		

Exercise 3

3–31 10-kg of R-134a fill a 1.348-m^3 rigid container at an initial temperature of −40°C. The container is then heated until the pressure is 200 kPa. Determine the final temperature and the initial pressure. *Answers:* 66.3°C, 51.25 kPa

Exercise 4

3–41 10-kg of R-134a at 300 kPa fills a rigid container whose volume is 14 L. Determine the temperature and total enthalpy in the container. The container is now heated until the pressure is 600 kPa. Determine the temperature and total enthalpy when the heating is completed.

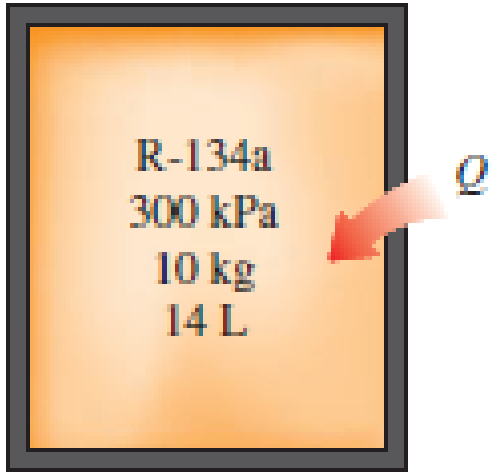


FIGURE P3–41

Exercise 5

3–42 100-kg of R-134a at 200 kPa are contained in a piston-cylinder device whose volume is 12.322 m^3. The piston is now moved until the volume is one-half its original size. This is done such that the pressure of the R-134a does not change. Determine the final temperature and the change in the total internal energy of the R-134a.

Exercise 6

3–43 Water initially at 200 kPa and 300°C is contained in a piston-cylinder device fitted with stops. The water is allowed to cool at constant pressure until it exists as a saturated vapor and the piston rests on the stops. Then the water continues to cool until the pressure is 100 kPa. On the T-v diagrams sketch, with respect to the saturation lines, the process curves passing through both the initial, intermediate, and final states of the water. Label the T, P and v values for end states on the process curves. Find the overall change in internal energy between the initial and final states per unit mass of water.

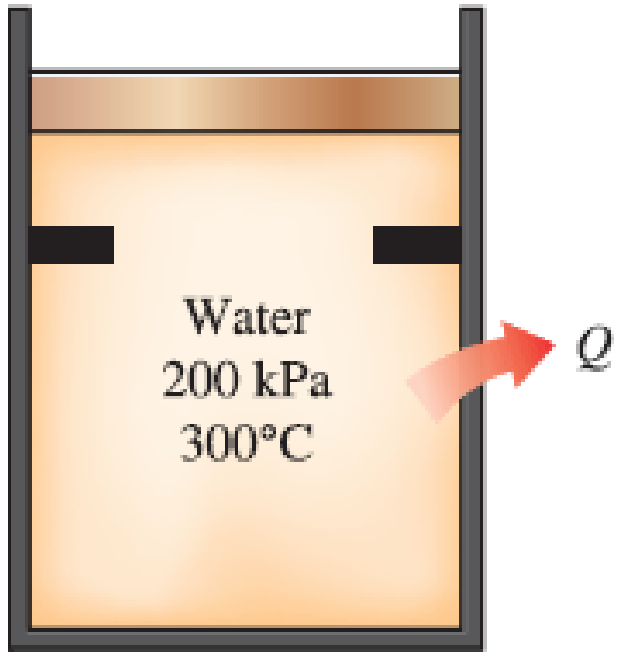


FIGURE P3–43

Exercise 7

3–64 A piston-cylinder device initially contains 50 L of liquid water at 40°C and 200 kPa. Heat is transferred to the water at constant pressure until the entire liquid is vaporized.

(*a*) What is the mass of the water?
(*b*) What is the final temperature?
(*c*) Determine the total enthalpy change.
(*d*) Show the process on a T-v diagram with respect to saturation lines.

Answers: (*a*) 49.61 kg, (*b*) 120.21°C, (*c*) 125,950 kJ

3.6 The Ideal-Gas Equation of State

- Gas and vapor are often used as synonymous words.
  - The vapor phase of a substance is customarily called a gas when it is above the critical temperature.
  - Vapor usually implies a gas that is not far from a state of condensation.
- Any equation that relates the pressure, temperature, and specific volume of a substance is called an equation of state.
- The simplest and best-known equation of state for substances in the gas phase is the **ideal-gas equation of state**, which predicts the *P-v-T* behavior of a gas quite accurately within some properly selected region.

$$Pv = RT; \qquad P = R\left(\frac{T}{v}\right)$$

Ideal gas equation of state

- *R* is called the **gas constant**, *P* is the **absolute pressure**, *T* is the **absolute temperature**, and *v* is the **specific volume**.

3.6 The Ideal-Gas Equation of State (2)

- The gas constant R is different for each gas and is determined from:

$$R = \frac{R_u}{M} \quad \text{(kJ/kg.K or kPa·m}^3\text{/kg·K)}$$

where R_u is the universal gas constant and M is the molar mass (also called molecular weight) of the gas.

Substance	R, kJ/kg·K
Air	0.2870
Helium	2.0769
Argon	0.2081
Nitrogen	0.2968

- The constant R_u is the same for all substances, shown below.
- The molar mass M is *the mass of one mole* (also called a gram-mole, abbreviated gmol) *of a substance in grams*, or the mass of one kmol (also called a *kilogram-mole*, abbreviated kgmol) in kilograms.
- The mass of a system is equal to the product of its molar mass M and the mole number N, i.e.

$$m = MN \quad \text{(kg)}$$

$$R_u = \begin{cases} 8.31447 \text{ kJ/kmol} \cdot \text{K} \\ 8.31447 \text{ kPa} \cdot \text{m}^3\text{/kmol} \cdot \text{K} \\ 0.0831447 \text{ bar} \cdot \text{m}^3\text{/kmol} \cdot \text{K} \\ 1.98588 \text{ Btu/lbmol} \cdot \text{R} \\ 10.7316 \text{ psia} \cdot \text{ft}^3\text{/lbmol} \cdot \text{R} \\ 1545.37 \text{ ft} \cdot \text{lbf/lbmol} \cdot \text{R} \end{cases}$$

3.6 The Ideal-Gas Equation of State (3)

- The ideal-gas equation of state can be written in several different forms:

$$V = mv \longrightarrow PV = mRT$$

$$mR = (MN)R = NR_u \longrightarrow PV = NR_uT$$

$$V = N\bar{v} \longrightarrow P\bar{v} = R_uT$$

  - where *v* is the molar specific volume, i.e., the volume per unit mole (in m^3/kmol).
  - A bar over property denotes values on a unit-mole basis.
- Writing the equation for two different states, $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$ Ideal gas equation at two states for a fixed mass
- An ideal gas is an imaginary substance that obeys the relation *Pv =RT.*
- In the range of practical interest, many familiar gases can be treated as ideal gas with negligible error (often < 1%)
  - Air, nitrogen, oxygen, hydrogen, helium, argon, neon, krypton, and even carbon dioxide (heavy gas)

3.6 The Ideal-Gas Equation of State (4)

Is Water Vapor an Ideal Gas?

- Fig. 3-46 shows the error involved in treating water vapor as an ideal gas.
  - At pressures below 10 kPa, water vapor can be treated as an ideal gas, regardless of its temperature, with negligible error (less than 0.1 percent).
  - At higher pressures, the assumption yields unacceptable errors, particularly in the vicinity of the critical point and the saturated vapor line.

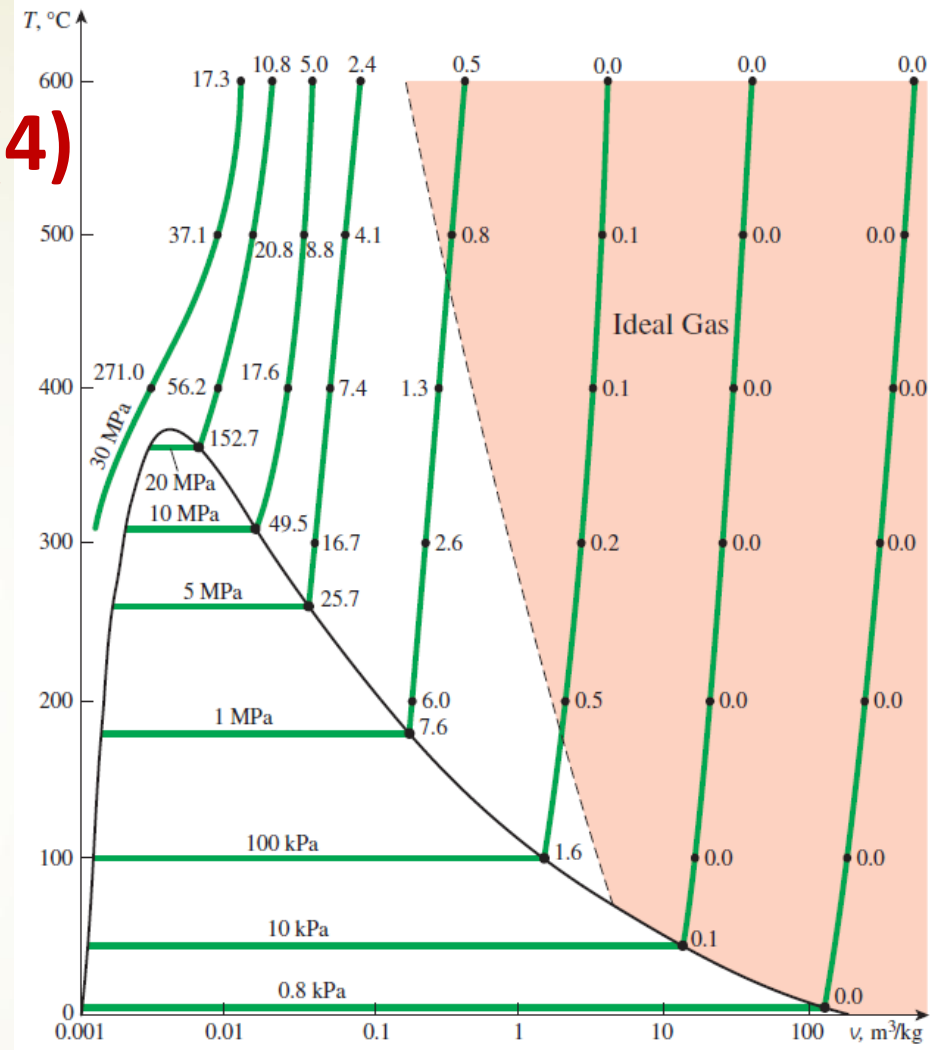


FIGURE 3–46
Percentage of error ($[|v_{table} - v_{ideal}|/v_{table}] \times 100$) involved in assuming steam to be an ideal gas, and the region where steam can be treated as an ideal gas with less than 1 percent error.

- Therefore, in air-conditioning applications, the water vapor in the air can be treated as an ideal gas with essentially no error since the pressure of the water vapor is very low.
- In steam power plant applications, however, the pressures involved are usually very high; therefore, the ideal-gas relation should not be used.

Exercise 6

3–77 Argon in the amount of 1.5 kg fills a 0.04-m^3 piston-cylinder device at 550 kPa. The piston is now moved by changing the weights until the volume is twice its original size. During this process, argon's temperature is maintained constant. Determine the final pressure in the device.

Exercise 7

3-78. A rigid tank contains 10 kg of air at 150 kPa and 20 °C. More air is added to the tank until the pressure and temperature rise to 250 kPa and 33 °C, respectively. Determine the amount of air added to the tank.

Review Exercise

3-107. One kg of water fills a 0.2825 m^3 weighted piston-cylinder device at a temperature of 350 °C. The piston cylinder device is now cooled until its temperature is 50 °C. Determine the final pressure and volume of the water. Also, show the process on P-v and T-v diagrams with respect to saturation lines.

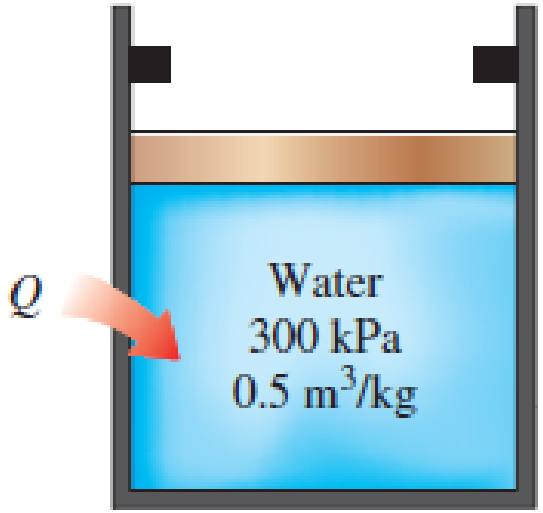
Review Exercise 2

3–127 If sufficient data are provided, complete the blank cells in the following table of properties of water. In the last column describe the condition of water as compressed liquid, saturated mixture, superheated vapor, or insufficient information; and, if applicable, give the quality.

P, kPa	T, °C	v, m³/kg	u, kJ/kg	Phase description
	250		2728.9	
300			1560.0	
101.42	100			
3000	180			

Review Exercise 3

3-128 Water initially at 300 kPa and 0.5 m^3/kg is contained in a piston-cylinder device fitted with stops so that the water supports the weight of the piston and the force of the atmosphere. The water is heated until it reaches the saturated vapor state and the piston rests against the stops. With the piston against the stops, the water is further heated until the pressure is 600 kPa. On the P-v and T-v diagrams sketch, with respect to the saturation lines, the process curves passing through both the initial and final states of the water. Label the states on the process as 1, 2, and 3. On both the P-v and T-v diagrams, sketch the isotherms passing through the states and show their values, in °C, on the isotherms.



Thank You